

Mohit Marvania

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RESEARCH INTERESTS

AI Safety and Robustness: Adversarial attacks on LLMs, Watermark resilience, and post-training attribution mechanisms.
Scalable Oversight: Verifiable reasoning, agentic planning benchmarks, and simulation-based stress-testing

EDUCATION

George Mason University <i>M.S. in Computer Science – Machine Learning</i> • Related Coursework: Artificial Intelligence, Data Mining, Machine Learning, Component-based Software Development, Advanced NLP.	Fairfax, Virginia <i>Expected, May 2026</i>
Charotar University of Science & Technology <i>Bachelor of Technology in Computer Science (GPA: 3.78/4.0)</i>	Changa, India <i>Graduated, April 2024</i>

SELECTED RESEARCH & PUBLICATIONS

Toward Resilient Watermark Detection: Stability-Aware Statistical Features

<i>Co-First Author</i> • Developed Pattern Stability Score (PSS), a novel statistical framework that solved a critical fragility in LLM watermarking where standard detectors collapse (AUC < 60%) under recursive paraphrasing attacks. • Achieved 94.6% AUC at depth-6 paraphrasing (vs. 66% baseline) and maintained greater than 81% AUC on ultra-short texts (300 tokens) where prior methods failed. • Architected a scalable O(n) complexity, detection pipeline on A100 clusters using Mistral-7B and Llama-2, optimizing the detection algorithm via rolling window analysis.	<i>Submitted to (ICML 2026)</i>
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TruthSim: A Controllable Framework for Realistic Medical AI Evaluation

<i>Co-First Author</i> • Designed a “Truth-Preserving” framework TruthSim , a patient simulator injecting 6 pillars of clinical noise (e.g. memory gaps, anxiety) to stress-test Medical/Doctor LLMs, utilizing UMLS-grounded verification to prevent hallucinations. • Revealed a critical “Sim-to-Real” gap: State-of-the-Art models (Llama-3-70B, Meditron) degraded by 15-25% diagnostic accuracy when facing realistic patient noise. • Conducted human evaluation and LLM-as-a-Judge studies (Cohen's Kappa=0.81 agreement), proving current static benchmarks overestimate medical AI robustness.	<i>Submitted to (ARR Jan 2026)</i>
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PROFESSIONAL EXPERIENCE

Graduate Research Assistant

<i>Supervisor: Dr. Abolfazl Safikhani & Dr. Ziwei Zhu, George Mason University.</i> • Adversarial Robustness (watermarking): Orchestrated large scale empirical evaluations using Mistral-7B, Llama-2, and Qwen to stress-test watermark resilience. Conducted rebuttal-phase experimentation, validating the PSS framework against novel attack vectors to ensure defense-in-depth. • Scalable Oversight & Reasoning: Evaluating reasoning capabilities of DeepSeek R1 (Distilled) & Llama-3 on constrained planning tasks. Developed a grid-based planning benchmark (3k+ maps) to test ‘deliberate prompting’ efficacy, achieving 80% near optimal routing. • Reproducibility: Implemented reproducible experiment pipelines with deterministic tokenization, seed control, and distributed metrics sweeping on Slurm-managed clusters.	Fairfax, VA <i>Jan 2025 – Present</i>
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Machine Learning Researcher Intern

<i>Supervisor: Dr. Amit Thakkar, Charusat University.</i> • Computer Vision Pipeline: Engineered a real-time engagement detection system (YOLOv8) achieving 72.3% accuracy for e-learning environments. Published findings in AIP Conference Proceedings (2025). • Dataset Creation: Automated the fusion of DaiSEE and YawDD datasets (4k+ samples) using OpenCV, reducing data preprocessing time by 80% via custom labeling scripts.	Changa, India <i>Dec 2023 – Apr 2024</i>
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Software Engineer Intern

<i>Raven Technolabs</i> • Optimized backend reliability for an AI-assisted receipt generation system serving 2000+ weekly users, reduced database write latency by 30% via batched operations and implemented CI/CD pipelines.	Rajkot, India <i>May 2023 – Jul 2023</i>
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SKILLS

Languages: Python (Advanced), C++, Java, SQL, Bash, JavaScript/TypeScript, Kotlin
ML Stack: PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas, Weights & Biases, DeepSpeed, RAG, Unsloth
Models: Llama 3, Mistral, DeepSeek R1, Qwen, GPT-4o API, Claude API.
Infrastructure: Docker, Kubernetes, Linux (Ubuntu), Slurm (Cluster Management), AWS (EC2 / S3).
Software Engineering: React.js, Next.js, HTML/CSS, FastAPI, Flask, REST, SQL, Pydantic, pytest, unit testing, code review. Core
Competencies: Content Authenticity, Large Scale Model Evaluation, Latency Optimization, Statistical Modeling.